

BUDAPEST PESTSZENTLORINC, Hungary

WMO: 128430

Lat: 47.433N

Lon: 19.183E

Elev: 456

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.4	(c) 18.1	(d) 6.1	(e) 7.7	(f) 20.4	(g) 10.1	(h) 9.4	(i) 21.7	(j) 19.9	(k) 41.6	(l) 17.4	(m) 40.1	(n) 3.6	(o) 90	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	92.0	69.4	88.8	68.3	85.7	67.2	71.5	85.8	70.1	84.3	68.7	82.3	6.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	101.2	76.8	65.5	95.8	75.2	64.0	90.7	74.2	35.5	85.6	34.2	84.7	33.1	82.3	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.4	(b) 14.0	(c) 12.0	(e) 9.5	(f) 96.3	(g) 5.4	(h) 3.6	(i) 5.6	(j) 98.8	(k) 2.5	(l) 100.9	(m) -0.6	(n) 102.9	(o) -4.5	(p) 105.5		
(a) 16.4	(b) 14.0	(c) 12.0	(e) 8.7	(f) 74.1	(g) 5.3	(h) 1.8	(i) 4.9	(j) 75.3	(k) 1.8	(l) 76.4	(m) -1.2	(n) 77.4	(o) -5.0	(p) 78.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	32.2	35.8	44.0	54.4	62.5	69.0	72.4	71.9	62.6	53.0	43.1	33.5
	DBStd	15.98	7.59	8.31	7.59	7.06	6.41	6.37	5.92	5.96	6.40	7.29	7.41	7.66
	HDD50	1997	553	401	213	37	1	0	0	0	0	53	226	513
	HDD65	5112	1018	819	652	321	124	31	9	11	120	373	656	978
	CDD50	3078	0	2	26	170	390	569	696	679	379	147	19	1
	CDD65	718	0	0	0	4	48	150	240	225	49	2	0	0
	CDH74	6762	0	0	0	61	457	1437	2319	2093	378	17	0	0
	CDH80	2434	0	0	0	6	93	501	927	824	83	0	0	0
Wind	WSAvg	5.5	5.2	5.7	6.3	6.1	5.7	5.5	5.7	5.0	5.1	4.9	5.3	5.1
Precipitation	PrecAvg	21.5	1.3	1.2	1.2	1.5	2.3	2.5	2.5	2.2	1.9	1.5	2.0	1.5
	PrecMax	36.1	3.5	4.8	4.4	4.1	6.4	6.7	7.2	8.0	5.7	5.9	6.4	4.1
	PrecMin	12.0	0.0	0.0	0.0	0.0	0.1	0.5	0.1	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.1	0.7	1.0	0.8	0.9	1.3	1.4	1.5	1.6	1.4	1.3	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	60.1	68.9	80.0	86.4	93.3	96.2	96.7	86.9	76.4	64.7	53.2
		MCWB	47.3	49.4	52.8	59.7	65.0	69.7	70.0	69.2	66.4	62.7	56.2	47.7
	2%	DB	48.8	55.1	64.1	75.1	82.6	89.6	91.9	92.0	82.1	72.3	60.1	48.9
		MCWB	43.8	46.8	50.6	57.3	63.8	68.9	69.7	68.9	64.8	60.4	53.5	45.2
	5%	DB	45.4	51.0	59.9	71.2	79.3	85.7	88.6	88.1	78.4	68.7	56.6	45.7
		MCWB	41.8	44.6	48.7	55.5	62.4	67.6	68.4	68.0	63.2	58.7	51.3	42.6
	10%	DB	42.4	47.4	56.2	67.6	75.7	82.2	85.3	84.5	74.8	64.8	53.6	43.4
		MCWB	39.4	42.7	46.8	53.7	60.9	65.9	67.3	66.9	61.6	56.7	49.5	40.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	50.1	54.7	61.3	67.3	72.9	73.5	73.5	68.3	63.9	56.7	48.7
		MCDB	53.4	58.1	65.3	76.2	82.0	86.7	88.8	87.2	81.5	73.7	62.3	51.8
	2%	WB	44.8	47.3	51.9	58.8	65.3	70.6	71.6	71.1	66.3	61.6	54.4	45.5
		MCDB	47.8	53.4	61.1	71.4	78.6	84.5	86.1	85.4	78.8	70.2	59.1	48.2
	5%	WB	42.3	45.4	49.9	56.8	63.8	69.1	70.1	69.6	64.6	59.6	52.1	43.3
		MCDB	45.0	49.9	58.1	68.3	76.0	82.5	84.1	84.4	75.2	66.8	55.8	45.4
	10%	WB	39.5	43.4	47.9	54.9	62.1	67.4	68.7	68.1	63.0	57.6	49.9	41.2
		MCDB	42.0	47.1	55.1	65.7	73.1	79.6	81.9	81.8	71.9	63.8	53.1	43.0
Mean Daily Temperature Range	5% DB	MDBR	9.4	12.1	15.3	18.0	18.5	18.8	19.3	19.5	17.2	16.0	11.1	8.4
		MCDBR	13.3	17.4	22.0	24.1	24.2	23.8	24.4	24.8	22.9	21.2	15.7	11.0
		MCWBR	9.2	11.1	12.1	10.8	9.8	8.4	7.8	7.9	9.3	10.7	9.8	8.5
	5% WB	MCDBR	12.8	15.5	19.4	21.8	21.7	21.5	21.4	22.1	19.8	19.5	14.4	10.4
		MCWBR	9.3	10.5	11.5	10.1	9.4	8.4	7.8	8.2	8.9	10.4	9.6	8.3
Clear-Sky Solar Irradiance	taub	0.330	0.353	0.390	0.439	0.428	0.429	0.442	0.441	0.416	0.384	0.359	0.329	
	taud	2.364	2.316	2.244	2.123	2.214	2.244	2.210	2.221	2.258	2.326	2.351	2.372	
	Ebn at Noon	234	253	260	258	266	266	260	254	249	238	220	219	
	Edh at Noon	23	30	37	46	43	43	43	41	36	29	23	21	
All-Sky Solar Radiation	RadAvg	329	588	983	1450	1750	1900	1861	1669	1174	752	397	250	
	RadStd	39	86	145	139	182	144	134	161	138	81	38	33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.10	N/A	N/A	N/A	+0.99	N/A	-189	-301	+189
(47) Regional (50 neighbors)		+1.06	N/A	N/A	N/A	+0.97	+0.97	-216	-315	+175

Nomenclature: See separate page